Sovereign Compliance Infrastructure for Regulated Finance

GPU-native compliance engine enabling cross-institutional financial crime detection without central data pooling. Every decision cryptographically attested. Every verdict independently verifiable.

150M+	29	6
POLICY EVALUATIONS / SEC	POLICIES IN PARALLEL	AML DOMAINS

The Problem

- **Siloed Detection:**
  Money laundering networks operate across banks. Individual institutions see fragments. An estimated 16 billion euros is laundered through the Netherlands alone each year. Less than 1% of illicit funds are intercepted.
- **Privacy vs. Detection:**
  Joint monitoring initiatives have stalled. Centralising transaction data triggers GDPR objections. The EU AMLR (effective mid-2027) restricts data sharing to pre-identified high-risk customers only.
- **Legacy Architecture:**
  95% false positive rates. Overnight batch processing on CPU rule engines. 50-100M+ euros annual compliance operations cost per Tier-1 bank. 5,500+ AML staff in the Netherlands alone.

The Solution

GPU-Native Processing

Full policy sets evaluated against every transaction simultaneously on GPU. Real-time, not batch. 150 million+ policy evaluations per second per installation.

Privacy-Preserving MPC

Cross-institutional detection via secure multi-party computation, validated in controlled testing with three simulated banks. No central data pool. No GDPR exposure. Each bank retains full data sovereignty.

Cryptographic Attestation

Every compliance decision produces an Ed25519-signed proof bundle with Merkle roots. Regulators verify independently with a standalone CLI tool.

Constitutional Governance

Zero-bypass enforcement. Every automated action requires a cryptographic warrant. Metered execution prevents runaway processes. Policy inclusion verifiable via Sparse Merkle Tree.

Deterministic Evaluation

Policy language compiled to GPU bytecode. Same input + same policy = same verdict. Always. Reproducible by any party at any time.

Entity Resolution at GPU Speed

GPU-resident identity graph with GNN-based risk propagation. Full network context for every decision in real time.

Core Engine

CAPABILITY	DETAIL
Processing speed	**150M+ CEPS. 500K entities in under 2 seconds. Alert lifecycle under 10ms.**
Policy language	**CPL: lexer, parser, semantic analyser, compiler to GPU bytecode**
Adjudication pipeline	**Triple-stream GPU: Load (H2D), Adjudicate (kernel), Commit (D2H + sign)**
Determinism	**Byte-identical replay guaranteed. Fixed-point arithmetic where applicable**
Entity resolution	**GPU-resident high-throughput hash table (~1.34GB VRAM), GNN risk scoring**
Proof generation	**ZAPB: verdict hash (SipHash-128), Merkle proof, Ed25519 signature (912 bytes)**
ZK governance proofs	**GPU-accelerated zero-knowledge governance proofs**

Governance & Safety

- Policy execution gateway: Zero-bypass enforcement point. No code path circumvents governance.
- Authorisation framework: Ephemeral Ed25519 keypairs. Scoped, time-limited, non-reusable action warrants.
- Resource governor: Monotonic per-agent gas ledger. Sharded concurrency. Prevents resource exhaustion.
- Predictive compliance engine: Forward simulation of compliance state. Predicts budget depletion, warrant expiry, and coverage gaps before they occur. Fixed-point deterministic projection.

Detection Intelligence

- Semantic Knowledge Graph: Multi-domain semantic encoding across entity existence, relationships, behaviour, epistemic state, and temporal projections. Contextual reasoning beyond raw transaction patterns.
- Multi-Timeline Evaluation: Same entity evaluated under multiple policy versions simultaneously on GPU. Scenario analysis, impact assessment, and counterfactual replay at compliance speed.
- Cross-Institutional MPC: Federation protocol: ECDH-PSI (X25519) for entity matching, Yao's Garbled Circuits (Free-XOR) for risk comparison, IKNP OT for oblivious transfer. Semi-honest security model. Under 10 seconds per bilateral round. Dual Ed25519 attestation. AES-256-GCM encrypted transport. Zero bytes of customer data shared.
- Financial Crime Network Simulator: GPU-native simulation of complete financial ecosystems with embedded criminal networks across 8 FATF-documented typologies. CBS-calibrated legitimate economy. Falsifiable, reproducible detection measurement. The industry benchmark that does not exist anywhere else.

Continuous Verification

- Adversarial testing framework: 13+ subsystems. GPU-accelerated fuzzing, decision surface cartography, adversarial oracle. Two-gate deployment pipeline (pre-commit + CI/CD).
- GapDetector: Real-time policy coverage mapping against registered regulatory frameworks. Gaps identified automatically, continuously.
- Build Attestation: BLAKE3 + SHA-256 + Ed25519 embedded in binary. Runtime self-verification. Compiler version and git commit recorded. Reproducible builds.

Integration

- Ingest: 256MB shared memory ring buffer. Lock-free atomic indices. JSON/REST, raw binary, GPU-Direct bypass. Configurable backpressure (drop newest, drop oldest, block).
- Export: RAW and CEF formats for SIEM (Splunk, QRadar, Sentinel). Zero-copy flat encoding. Domain-tagged BLAKE3 integrity. Every exported verdict traceable to its proof bundle.

Market Opportunity

- 200B+ euros annual global compliance spending. Growing 15-20% per year. Banks allocate 10-20% of operating budgets to compliance.
- AMLR mid-2027 creates a hard regulatory deadline. Every European bank must re-architect. EU AI Act classifies AML as high-risk with mandatory technical standards from August 2026.
- Cross-institutional detection proven valuable (TMNL demonstrated this over 4 years) but centralised architecture failed on privacy. MPC-based alternative has no competitors.

Competitive Landscape

	LEGACY INCUMBENTS	AI-NATIVE CHALLENGERS	ZQUAS
Examples	NICE Actimize, Oracle, SAS	Sardine ($145M), Hawk ($56M)	---
Architecture	CPU batch, rules-based	CPU cloud, ML-based	**GPU-native, MPC + ZK**
Cross-bank detection	No	No	**Yes (validated, pilot-ready)**
Cryptographic proof	No	No	**Yes (Ed25519, Merkle)**
Deterministic eval	No	No	**Yes (byte-identical)**
Real-time at scale	Limited	Partial	**150M+ CEPS. Under 2s at 500K entities.**
Adversarial self-testing	No	No	**Continuous (GPU)**
Regulator verification	Trust-based	Trust-based	**Independent CLI**
Replication timeline	N/A	N/A	**3-5 years minimum**

Traction

- DNB InnovationHub: Submission under review
- FCA Sandbox: Regulatory sandbox programme engaged
- NVIDIA Inception program member
- Benchmark: 150M+ CEPS on RTX 5090. 500,000 entities in under 2 seconds. Alert lifecycle under 10ms.
- Cross-institutional detection: under 10 seconds per bilateral round. 4/4 typologies detected.
- Test coverage: 7,200+ automated tests across 12 audited subsystems
- Regulatory frameworks: EU AI Act, AMLR, FATF R15, NIST AI RMF, ISO 42001, MAS TRM, GDPR, DORA

Business Model

Land:

Single-bank on-premise deployment. Annual licensing by transaction volume and policy complexity. Bank's data never leaves its infrastructure.

Expand:

Additional business lines, policy domains, jurisdictions within existing banks. Net revenue retention above 100%.

Network:

Cross-institutional MPC activates as the federation grows. Each new bank increases detection for all participants. Switching costs deepen with cryptographic audit chain.

How Cross-Institutional Detection Works

Banks already monitor transactions. Criminals exploit the gaps between banks. ZQUAS adds a federation layer that makes cross-institutional patterns visible without sharing any data. Each bank keeps its existing systems and processes.

What Single-Bank Monitoring Misses

A criminal entity has accounts at three banks. At each bank, the transaction pattern looks normal. Across all three, the pattern is classic money laundering: money enters through one bank, fragments through a second, and reconsolidates through a third. No single bank can see this.

From Transaction to SAR in Six Steps

- **1. Local Monitoring**
  Banks monitor transactions using their existing systems. Risk scores per entity based on local patterns. Unchanged.
- **2. Federation Round**
  ZQUAS installations execute bilateral rounds (A-B, A-C, B-C). Private Set Intersection reveals shared entities. Secure comparison checks combined risk against threshold. Neither bank sees the other's score. Ed25519 attestation per round.
- **3. Escalation Alert**
  If combined risk exceeds threshold, both banks receive an alert: entity ID, escalation flag, attestation proof. No transaction data. No individual scores. Delivered to existing case management via API or file import.
- **4. Enhanced Due Diligence**
  Analyst receives the alert in their normal workflow. Reviews their own transactions. The new information: this entity has elevated cross-institutional risk.
- **5. SAR Decision**
  Each bank independently decides whether to file with FIU-Nederland. The bank uses only its own data in the SAR. The cross-institutional alert is the trigger, not the content.
- **6. FIU Investigation**
  FIU receives separate SARs from multiple banks about the same entity. The complete laundering pattern becomes visible. No single bank could have seen it alone.

Measured on Production Hardware (RTX 5090)

METRIC	RESULT
Single-bank pipeline (500K entities, 100 policies)	**Under 2 seconds**
Bilateral federation round (100K entities per bank)	**Under 10 seconds**
Peak throughput	**150M+ CEPS (compliance policy evaluations/sec)**
Typologies detected	**4/4 at every scale, per-entity verified**
Transport security	**AES-256-GCM + Ed25519 on every round**
Data shared between banks	**Zero bytes of customer data**

Four real-world laundering typologies tested: trade-based money laundering, correspondent banking wire stripping, shell company layering, and funnel account structuring. All four detected. Zero bytes of customer data shared between banks.

What Each Bank Learns

One new fact per shared entity: whether the combined risk exceeds the agreed threshold. Not the other bank's score. Not their transactions. One bit of information. Enough to trigger investigation, not enough to compromise privacy.

Founder

Danny de Gier Founder & Engineer

Compliance Career

18+ years in financial crime compliance at Tier-1 banks and fintechs. Senior roles at RBS, Deutsche Bank, HSBC, and Commerzbank. Fintech compliance leadership at ClearBank, Vivid Money, and CoinMetro. Professional Postgraduate Diploma in Financial Crime Compliance (International Compliance Association / University of Manchester). Has personally built monitoring frameworks, written SARs, sat through regulatory examinations, and managed compliance teams across multiple jurisdictions.

Engineering

Self-taught GPU systems programmer. C++23, CUDA (sm_86/89/100/120), Vulkan. Built the entire ZQUAS engine as a solo developer: policy language compiler, GPU adjudication kernel, MPC protocol implementation, cryptographic attestation pipeline, adversarial testing framework, entity resolution graph, rendering engine. This dual expertise, compliance domain knowledge combined with GPU systems programming, shapes every architectural decision and is extremely rare in the talent market.

Engineering Standards

STANDARD	DETAIL
Language	C++23 with CUDA
Codebase	396,000 lines of C++ and CUDA, 493 GPU kernels
GPU targets	sm_86, sm_89, sm_100, sm_120
Build hardening	/sdl, /guard:cf, /GS, /Qspectre, /CETCOMPAT, /HIGHENTROPYVA
Testing	7,200+ automated tests across 12 audited subsystems
Cryptography	BLAKE3, SHA-256, Ed25519, ZK-optimised hash functions, SipHash-128
Entity system	High-performance entity component system
Serialisation	Custom binary format (64B-aligned, std140-compatible)
Build attestation	BLAKE3 + SHA-256 + Ed25519 embedded, reproducible builds

Regulatory Framework Coverage

EU AI Act | AMLR / 6AMLD | FATF R15 | NIST AI RMF | ISO 42001 | MAS TRM | GDPR | DORA

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The architecture is ready. The regulatory window is open. Let's talk.

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